

Syn Bank Share of Wallet Intelligence Engine

Team: Root3

Luke Naidoo • Wisdom Ejiro Peru • Faten Saud

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20

JSE-listed clients
analyzed (43 company-years)

5.8%

avg. blended Share of
Wallet, actionable-tier clients

R2.4tn

combined addressable gap,
10 highest-confidence clients

0.24–0.39

ElasticNet R², 4/5 targets
(honest LOGO
cross-validation)

3/3

GenAI use cases built:
briefing notes, NL query,
anomalies

THE PROBLEM

Syn Bank is never a client's **sole** banking partner. Our team built an engine that estimates each client's **Total Wallet** across three product pillars, quantifies Syn Bank's **current Share** from internal data, and ranks the **Rand Gap** to prioritize which clients SynBank should chase next — for the 20 JSE-listed corporates in Syn Bank's synthetic transactional, trade finance, and cross-border datasets.

OUR APPROACH

Numerator (Share): Internal transactional, trade finance and cross-border data, split into 3 pillars — Transactional Banking, Trade & Working Capital, Foreign/Cross-Border.

Denominator (Wallet): top-down proxy built from each client's own disclosed annual-report financials (revenue, cost of sales, receivables/payables, foreign-revenue %), extracted from 43 company-year filings (FY2024–2026) and normalized to ZAR at live SARB rates.

Learned angle: ElasticNet regression (5 targets, Leave-One-Group-Out CV) predicts share % directly from the internal activity volume — useful precisely where the top-down external benchmark is thin or missing (when the company's public financial statements do not disclose enough reliable information to construct the external top-down wallet estimate).

GENAI LAYER (3/3 USE CASES)

Client briefing notes — Converts each client's Share of Wallet, estimated walletgaps, reliability indicators and model signals into a concise briefing, helping relationship managers quickly understand where the biggest opportunities lie and what to discuss with the client.

Anomaly explanations — Automatically identifies unusual or high-opportunity wallet signals and uses GenAI to explain why they matter, while keeping the underlying detection and calculations deterministic to reduce hallucination risk.

NL query assistant — Lets bankers interrogate the portfolio in plain English, asking questions such as "Which three clients should we prioritise?" or "Where do the ML and financial-statement estimates disagree?" The assistant grounds its answers in both the top-down financial benchmark and the latest ElasticNet predictions.

KEY FINDINGS

- **Top opportunity:** Shoprite Holdings — 2.4% transactional share, R437bn gap on a R448bn wallet (literal, ZAR-reporter).
- **Sharpest anomaly:** Valterra Platinum — near-zero share (0.0–0.2%) despite being a *domestic* reporter, not a scale artifact — flagged for immediate relationship review.
- **ML performance:** after target-specific ElasticNet hyperparameter tuning, out-of-sample R² reached 0.394 for transactional revenue, 0.340 for trade receivables, 0.269 for transactional cost, and 0.241 for trade payables. FX remained weak (R² = –0.264), reflecting the particularly sparse 17-observation FX training set.
- **Model self-awareness:** reliability tiering catches the naive-model trap before it reaches a banker — a raw calculation would report Glencore's global consolidated revenue as a fictional R9.5 trillion "gap"; explicitly flagged low-reliability/directional-only instead.

LIMITATIONS, STATED PLAINLY

- **Limited ML training sample:** only 43 company-year observations were available across 20 corporates. The original FY2025-only scope (~20 observations) was therefore expanded to FY2024–2026 where available, more than doubling the training data.
- **FX remains the weakest target:** only 17 usable FX observations remained after filtering, resulting in weak out-of-sample performance (R² = –0.264). FX estimates are therefore treated as directional rather than precise.
- **Designed around the constraint:** ElasticNet regularisation, company-level Leave-One-Group-Out cross-validation and target-specific hyperparameter tuning were used to reduce overfitting and improve generalisation within the available sample.

NEXT STEPS

- Layer in bottom-up competitor evidence (SENS facility announcements, BA900) to tighten/validate the top-down proxy.
- Expand the multi-year training dataset, particularly Foreign/Cross-Border observations, to improve model generalisation and strengthen the currently weak FX prediction.
- Automate the GenAI briefing-note pipeline via API to scale from 20 to the full 43-row multi-year panel.

Reliability tiers, stated on every figure:

10 moderate (ZAR reporter, literal Rand gap)

9 low (foreign currency, directional % only)

1 insufficient (Group AFS not disclosed)